

Data Scientist Challenge: Enterprise RAG with Verifiable Citations

Time Limit: 3 days **Deliverable:** .zip file

1. Business Problem Statement

Context: You have just joined PolicyPal AI, an internal AI initiative aimed at enhancing HR operations within Omnicorps Inc. The goal is to empower employees and HR specialists with instant, accurate answers to common policy questions, freeing HR staff to focus on more complex, human-centric work.

Example questions may include:

- “What services and vacation is a bereaved employee entitled to?”
- “How many days can I work remotely?”

However, there is a significant challenge: the HR team does not fully trust the AI’s responses. HR specialists often need to manually verify the information provided by the AI bot, which defeats the purpose of automation and slows down the process.

The Problem: Current systems typically provide a synthesized answer but lack the transparency and verifiability that HR requires. For example, when the bot states, “You get 20 vacation days,” the HR specialist needs to quickly see exactly where that information appears in the official Employee Handbook. Without direct and granular supporting evidence, trust in the AI remains low. This results in continued manual verification and limits the efficiency gains expected from automation.

Your Challenge: Your task is to improve the system in a way that increases HR’s confidence in the AI’s responses. You will develop an improved solution from scratch while addressing the limitations of the current system.

The Data: You will be provided with a copy of Omnicorps' comprehensive "Employee Handbook" in PDF format (e.g., Omnicorps_Employee_Handbook.pdf).

Use of AI Tools: You are permitted to use AI tools, including coding assistants and large language models, to support your work. However, you must be able to clearly explain your solution. The reference section of the solution README should include:

1. The AI tool(s) used.
2. The purpose for which each tool was used, such as brainstorming, coding, debugging, documentation, test case generation, or explanation.
3. Any known limitations, assumptions, or concerns related to the AI-assisted content.

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Submission:

Submit your solution as a .zip file. You may use any repository or folder structure that you consider appropriate.

The final solution must be orchestrated using a Jupyter Notebook, and the notebook should be submitted with all output cells included. You can use separate Python modules, helper functions, configuration files, prompts, supporting scripts, etc. as part of your solution. The notebook should serve as the main entry point that demonstrates and runs the end-to-end workflow. Use markdown cells in the notebook, where appropriate, to explain your approach. You should also include a README file to provide additional supporting information.